

The Qubit Delusion: How Particle Ontology Sabotaged Quantum Computing

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Abstract

The quantum computing industry has absorbed approximately \$35 billion in global investment over two decades while delivering zero commercially viable machines. This paper argues that the failure is not merely an engineering delay but an epistemic crisis: the dominant qubit-gate-circuit model of quantum computation imports a particle ontology that is inconsistent with quantum field theory, relational quantum mechanics, and the physics of continuous, correlated systems. Through systematic deconstruction of the scaffolds underlying quantum computing – qubit-as-particle, gate-as-discrete-operation, decoherence-as-enemy, error-correction-as-classical-coding – we identify the invariants that survive scaffold removal (superposition, entanglement, correlation structure, phase coherence) and examine alternative computational paradigms (measurement-based, continuous-variable, topological, and field-theoretic) that are more faithful to quantum reality. We combine ontological analysis with empirical forensics – a reproducibility scorecard of twenty major milestone claims, a comparative bubble-morphology analysis situating the quantum computing capital structure alongside historical technology bubbles, and a press-release-versus-preprint claim-gap analysis – to assess whether the field is producing science or narrative assets. The evidence suggests a systematic map-territory confusion operating at industrial scale, sustained by institutional incentives that reward optimism over falsification. We conclude with a constructive research agenda grounded in field-theoretic and relational quantum computation.

1. Introduction

In October 2019, Google announced “quantum supremacy” — a 53-qubit processor, Sycamore, had performed a random circuit sampling task in 200 seconds that would, Google claimed, take the world’s largest supercomputer 10,000 years to reproduce [Arute2019]. Sundar Pichai compared the achievement to the Wright brothers’ first flight. The announcement triggered a wave of media coverage, venture capital inflows, and national quantum strategies across the G20.

Four years later, IBM’s Head of Quantum, Jay Gambetta, reframed the field’s ambitions. Gone was “supremacy.” The new watchword was “quantum utility” — a more modest claim that quantum computers might someday solve problems of practical interest. IBM’s Eagle processor, with 127 qubits, had simulated the Ising model on a scale that strained classical methods [Kim2023]. Yet no commercially valuable computation had been performed that could not, with some effort, be accomplished on classical hardware.

This pattern — spectacular announcement followed by qualification, followed by a new, more distant milestone — has repeated with metronomic regularity throughout quantum computing’s commercial history. D-Wave sold “quantum computers” to Lockheed Martin and Google in 2011

– 2013, only for independent benchmarking to reveal no speedup over classical algorithms for the problems tested [Ronnow2014]. Rigetti went public via SPAC at a \$1.5 billion valuation in 2022; by 2024 its stock had declined over 90%. IonQ, the first pure-play quantum computing company to IPO, has oscillated between hype-driven rallies and reality-driven corrections.

By 2026, the global quantum computing industry has absorbed an estimated \$35 billion in combined public and private investment. No commercially viable quantum computer exists. Not one. The roadmaps project fault-tolerant machines in the 2030s, which would mean a roughly forty-year interval between Shor’s algorithm (1994) — the field’s foundational existence proof — and a working product. Even those timelines assume flawless execution of error-correction protocols that currently require thousands of physical qubits to encode a single logical qubit.

The standard narrative attributes this protracted timeline to engineering difficulty: qubits are fragile, decoherence is relentless, error correction is expensive, and scaling is hard. This is undoubtedly true. But it may also be *insufficient* — a proximate explanation that obscures a deeper, structural problem.

This paper proposes a different diagnosis. **The failure of quantum computing to produce viable hardware is not primarily an engineering problem. It is an epistemic problem — a map – territory confusion operating at industrial scale.** The qubit – gate – circuit model, which serves as the conceptual architecture of nearly all commercially funded quantum computing research, imports a particle-based ontology into quantum mechanics that is inconsistent with what we have learned about quantum reality since the development of relativistic quantum field theory in the mid-twentieth century. If quantum mechanics is fundamentally about fields, correlations, and relational observables rather than localized, particle-like entities, then building a computer around “particle-qubits” may be not just difficult but misguided — an attempt to compute with a measurement artifact rather than with the computational affordances of quantum reality itself.

This paper develops this thesis in five parts. **Section 2** provides a philosophical and physical deconstruction of the qubit – gate – circuit scaffold, identifying what is arbitrary convention (scaffold) and what is invariant across representations. **Section 3** presents empirical forensics: a reproducibility scorecard for twenty major quantum computing milestone claims, a bubble-morphology analysis comparing the quantum computing capital structure to historical technology bubbles, and a press-release-versus-preprint claim-gap analysis across five leading institutions. **Section 4** surveys alternative quantum computational paradigms — measurement-based, continuous-variable, topological, and field-theoretic — that move away from particle-qubit ontology and may be more faithful to quantum reality. **Section 5** examines the sociology and institutional economics of the quantum computing ecosystem, analyzing how career incentives, venture capital structures, and corporate branding imperatives sustain the qubit-gate model even in the face of mounting negative evidence. **Section 6** proposes a constructive research agenda grounded in field-theoretic and relational approaches to quantum computation. **Section 7** concludes.

A note on method: this paper is produced by the QNFO autonomous research pipeline under the LLM Research Automation Protocol (LRAP). All empirical claims are sourced from publicly available literature, press archives, preprint servers, and financial disclosures. The ontological analysis draws on the Deconstruction Spiral v4.0 methodology, which systematically separates human-imposed scaffolds from structural invariants. The empirical forensics employ standard meta-scientific tools adapted from the reproducibility movements in psychology and biomedicine. No proprietary data or human-subject research is involved. The conclusions are falsifiable: Section 3 specifies the observations that would refute the central thesis.

2. Ontological Deconstruction of the Qubit – Gate – Circuit Model

2.1 The Scaffold – Invariant Distinction

Every scientific model deploys *scaffolds*: arbitrary human conventions — choice of basis, coordinate system, unit, epoch, gate set, qubit encoding — that enable representation but do not inhere in the phenomenon represented. The ability to distinguish scaffold from invariant is the mark of episte